11) Experiment to understand KNN Algorithm for a given data set.

Program:

```
import numpy as np
import pandas as pd
df = pd.read_csv('/Iris (1) - Iris (1).xlsx - Iris (1) - Iris (1).csv')
df.info()
df.variety.value_counts()
df.head()
features = df.iloc[:, :-1].values
label = df.iloc[:, 4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain, xtest, ytrain, ytest = train_test_split(
    features, label, test_size=0.2, random_state=0
)
model_KNN = KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain, ytrain)
model_KNN.score(xtest, ytest)
model_KNN.score(xtrain, ytrain)
from sklearn.metrics import confusion_matrix, classification_report
ypred = model_KNN.predict(xtest)
print(confusion_matrix(ytest, ypred))
print(classification_report(ytest, ypred))
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
[[11  0  0]
 [ 0 12  1]
 [ 0  0  6]]
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        11
  Versicolor       1.00      0.92      0.96        13
   Virginica       0.86      1.00      0.92         6

    accuracy                           0.97        30
   macro avg       0.95      0.97      0.96        30
weighted avg       0.97      0.97      0.97        30
```

Result:

Experiment is conducted to understand the KNN Algorithm for the given data set.